

lab4_hw

AUTHOR

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```
library(LexisNexisTools)
library(quanteda)
library(tm)
library(topicmodels)
library(topicdoc)
library(tidyverse)
library(tidytext)
library(reshape2)
library(tictoc)
library(LDAvis)
library(tsne)
library(servr)
```

Load in Data

```
#tbl <- read_csv(here::here("week5", "Nexis", "ocean-acidification"))

pre_files <- list.files(
  path = here::here("week5", "Nexis", "ocean-acidification"), # update to your folder
  pattern = "\\\\.docx$",
  full.names = TRUE,
  recursive = TRUE,
  ignore.case = TRUE
)

pre_dat <- lnt_read(pre_files)

pre_meta_df <- pre_dat@meta
pre_articles_df <- pre_dat@articles
pre_paragraphs_df <- pre_dat@paragraphs

df <- tibble(
  date = pre_meta_df$Date,
  headline = pre_meta_df$Headline,
  id = pre_meta_df$ID,
  text = pre_articles_df$Article,
  source = pre_meta_df$Newspaper
)
```

1. Build and clean my corpus

Build Corpus

```
tbl <- df
corpus <- corpus(x = tbl,
                 textg_field = "text")
```

Tokenize and Clean

```
# Stop words
add_stops <- c(quanteda::stopwords("en",
                                   source="smart"),
              # remove other meta data words
              "said", "says", "told", "according", "discuss", "because", "where", "vol",

# Tokenize
toks <- tokens(corpus,
               remove_punct = TRUE,
               remove_numbers = TRUE,
               remove_symbols = TRUE) %>%

# Normalize
tokens_tolower()

# Remove stop words
toks1 <- tokens_select(toks, pattern = add_stops, selection = "remove")
```

Build the Document-Feature Matrix

```
# Creating a dfms
dfm1 <- dfm(toks1)
dfm2 <- dfm_trim(dfm1, min_docfreq = 2) # Reduce by limiting min values of feature's doc

# Remove empty rows b/c LDA will error on these
sel_idx <- slam::row_sums(dfm2) > 0 # select no empty rows
dfm <- dfm2[sel_idx,] # subset
```

2. Select K using multiple criteria

Step 1: Initial Model (Theory-Driven k)

Run 3 models (K = 3)

```
# Define Hyperparameter
k <- 3
```

```
# Model
topicModel_k3 <- LDA(dfm, # matrix
                     k,
                     method = "Gibbs", # Gibbs is default (always do this)
                     control = list(iter = 100, # Number of iterations
                                     verbose = 25, # Tell us after every 25 iterations
                                     seed = 321
                                   ))
```

```
K = 3; V = 11186; M = 159
Sampling 100 iterations!
Iteration 25 ...
Iteration 50 ...
Iteration 75 ...
Iteration 100 ...
Gibbs sampling completed!
```

Inspect the Posterior Distributions

```
result <- posterior(topicModel_k3)

beta <- result$terms
theta <- result$topcis

cat("phi dimensions (topics × vocab):", dim(beta), "\n")
```

```
phi dimensions (topics × vocab): 3 11186
```

```
# Top 10 terms per topic
terms(topicModel_k3, 10)
```

	Topic 1	Topic 2	Topic 3
[1,]	"climate"	"ocean"	"coral"
[2,]	"al"	"coral"	"reef"
[3,]	"access"	"marine"	"fig"
[4,]	"open"	"reefs"	"species"
[5,]	"change"	"corals"	"carbonate"
[6,]	"research"	"ecosystems"	"fish"
[7,]	"global"	"sea"	"corals"
[8,]	"energy"	"global"	"reefs"
[9,]	"carbon"	"climate"	"data"
[10,]	"environmental"	"bleaching"	"restoration"

K=3 Model Results/Groups:

There are 3 topics (k=3):

1. climate, al, access, open, change

2. ocean, coreal, marine, reefs, corals

3. coral, reef, fig, species, carbonate

(The first words in each list are the most probable word for each topic /highest probability of word occurrence)

Step 2: Selecting k with Coherence Scores

To select the best K, will run the model over a range of Ks. The coherence scores measure how related the words within a generated topic. Therefore, the score will give use a good idea on an optimal K.

```
# Fit model for a range of k values
k_values <- seq(2, 15, by = 1)

tic() # Timerstart

# Running the model
coherence_results <- map_dfr(k_values, function(k){ # Loop thru Ks
  # Run the model
  m <- LDA(dfm, k = k,
    method = "Gibbs",
    control = list(iter = 300,
      seed = 231,
      verbos = 0))
  # Calculate coherence
  coh <- mean(topic_coherence(m, dfm))
  tibble(k=k, coherence = coh)
})

toc() # Timer end
```

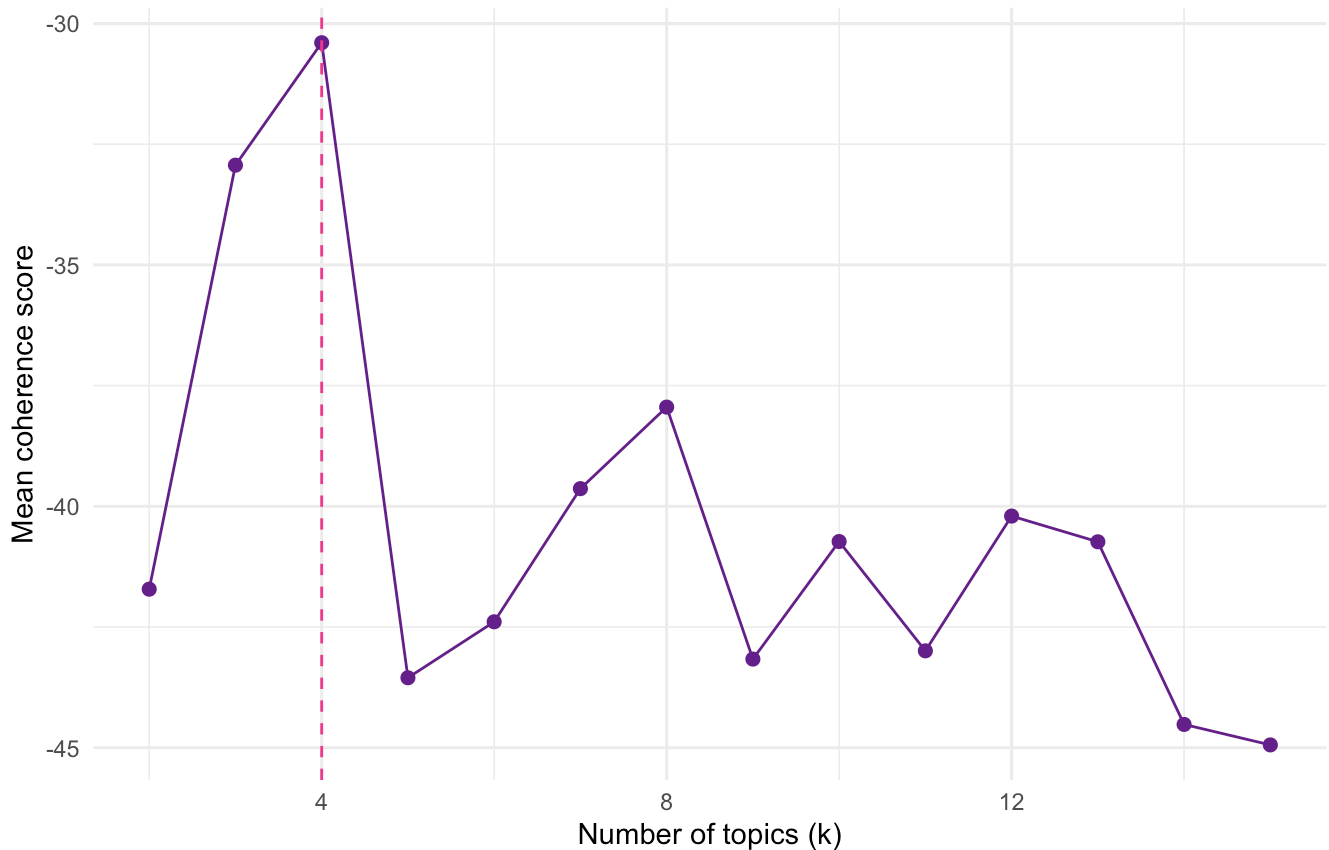
42.511 sec elapsed

Now, lets plot the coherence vs K!

```
ggplot(coherence_results, aes(x = k, y = coherence)) +
  geom_line(color = "darkorchid4") +
  geom_point(color = "darkorchid4", size = 2) +
  geom_vline(xintercept = coherence_results$k[which.max(coherence_results$coherence)], #
    linetype = "dashed", color = "violetred2") +
  labs(title = "Mean topic coherence by number of topics (k)",
    subtitle = "Dashed line = k with highest coherence",
    x = "Number of topics (k)",
    y = "Mean coherence score") +
  theme_minimal()
```

Mean topic coherence by number of topics (k)

Dashed line = k with highest coherence



Optimization metrics: The peak coherence (the “elbow”) is at $K = 4$ with a K of 3 following closely behind. Interestingly, a K of 5 drops significantly, meaning the region is unstable.

Theory: Possible ocean acidification themes: coral reef biology, ocean chemistry, climate policy/governance, and marine organism impact.

Interpretability: With a $K = 3$ (from Step 1), topics 2 and 3 are practically the same groupings. Due to this heavy overlap, $K = 3$ is under-splitting meaning a K of 4 could be favorable.

The coherence score shows $K = 4$ being favorable. Four ocean acidification themes come to mind. Lastly, step 1's initial model with a K of 3 is under-splitting. Thus, a K of 4 seems to be optimal.

3. Fit your final model and visualize top terms

Step 3: Fit the Selected Model

```
# Updated with the peak coherence K (k = 5)
k <- 4

topicModel_k_select <- LDA(dfm, k,
                           method = "Gibbs",
```

```
control = list(iter = 1000,
               verbose = 25,
               seed = 321))
```

Show the 4 topics:

```
tmResult <- posterior(topicModel_k_select)
terms(topicModel_k_select, 10)
```

	Topic 1	Topic 2	Topic 3	Topic 4
[1,]	"climate"	"coral"	"coral"	"climate"
[2,]	"ocean"	"reef"	"reefs"	"al"
[3,]	"marine"	"fig"	"corals"	"access"
[4,]	"change"	"carbonate"	"ocean"	"open"
[5,]	"sea"	"species"	"bleaching"	"change"
[6,]	"warming"	"corals"	"reef"	"research"
[7,]	"coastal"	"fish"	"marine"	"energy"
[8,]	"global"	"data"	"species"	"global"
[9,]	"ecosystems"	"reefs"	"scientists"	"carbon"
[10,]	"areas"	"analysis"	"world"	"pdf"

```
# Values per topic per text
theta <- tmResult$topics
# Values per vocab word
beta <- tmResult$terms
# Vocab words
vocab <- colnames(beta)
```

Step 4: Visualize Top Terms per Topic

Visualizing the 5 topics and the most probable words for each topic.

```
topics <- tidy(topicModel_k_select, matrix = "beta")

top_terms <- topics |>
  group_by(topic) |>
  slice_max(beta, n = 10) |>
  ungroup() |>
  arrange(topic, -beta)

top_terms |>
  mutate(term = reorder_within(term, beta, topic, sep = "")) |>
  ggplot(aes(term, beta, fill = factor(topic))) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~ topic, scales = "free_y") +
  scale_x_reordered() +
  coord_flip() +
  labs(title = "Top 10 terms per topic",
```

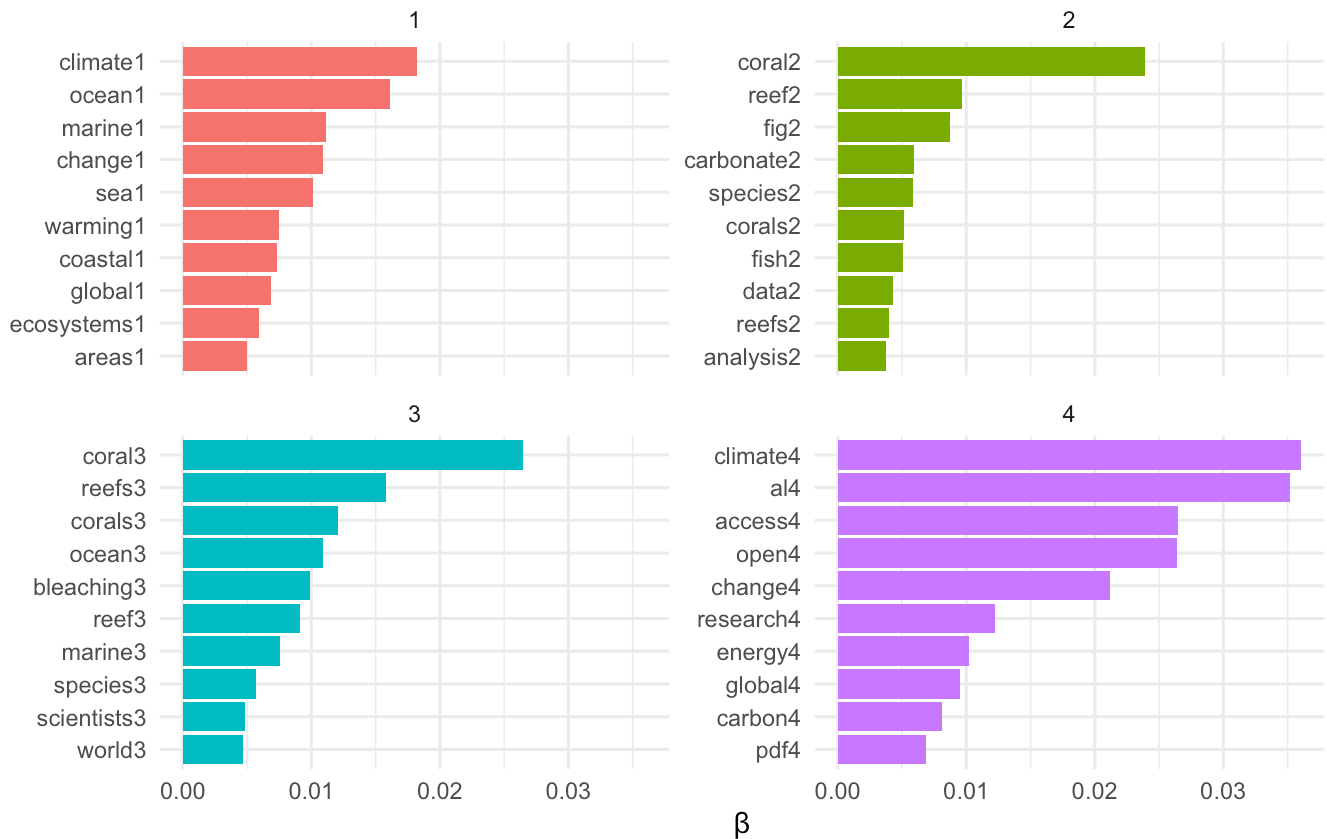
```

  subtitle = "β = P(term | topic)",
  x = NULL, y = "β") +
  theme_minimal()

```

Top 10 terms per topic

$\beta = P(\text{term} | \text{topic})$



Step 5: Name the Topics

Naming the topics

Assign a descriptive name to each topic (not just the concatenated top terms).

```

# Option 1: auto-name from top 5 terms
# topic_words <- terms(topicModel_k_select, 5)
# topic_names <- apply(topic_words, 2, paste, collapse = " ")

# Option 2: Replace topic names with interpretive labels after inspection
topic_names <- c("climate_change", "coral_reef_ecology",
                 "reef_biodiversity", "research")

topic_names

```

```

[1] "climate_change"      "coral_reef_ecology" "reef_biodiversity"
[4] "research"

```

Step 6: Document-Topic Distributions (θ)

Distribution of topics on Documents

Plot θ for the first 6 document.

```
# Grabbing first 6 documents
example_ids <- c(1:6)
n <- length(example_ids)

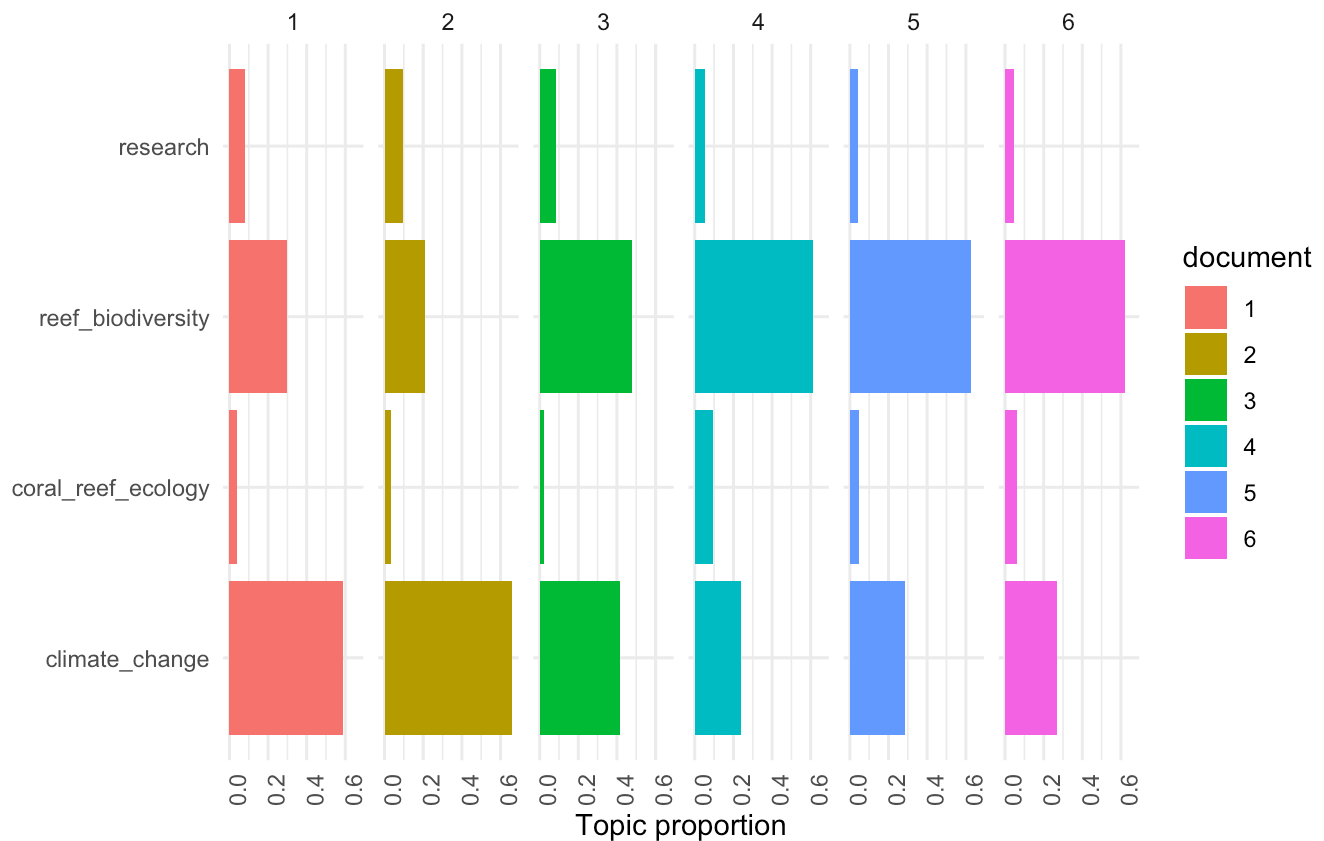
example_props <- theta[example_ids, ]
colnames(example_props) <- topic_names

# Value per topic per document
viz_df <- melt(cbind(data.frame(example_props),
                             document = factor(1:n)),
              variable.name = "topic",
              id.vars = "document")

# Plot
ggplot(data = viz_df, aes(topic, value, fill = document)) +
  geom_bar(stat = "identity") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  coord_flip() +
  facet_wrap(~ document, ncol = n) +
  labs(title = "Topic distributions across sample documents",
       subtitle = " $\theta = P(\text{topic} \mid \text{document})$ ",
       x = NULL, y = "Topic proportion")
```


Topic distributions across sample documents

$\theta = P(\text{topic} \mid \text{document})$



Step 7: Interactive Visualization with LDAvis

```
svd_tsne <- function(x) tsne(svd(x)$u)

json <- createJSON(
  phi      = tmResult$terms,
  theta    = tmResult$topics,
  doc.length = rowSums(dfm),
  vocab     = colnames(dfm),
  term.frequency = colSums(dfm),
  mds.method = svd_tsne,
  plot.opts  = list(xlab = "", ylab = "")
)

#serVis(json)
```

LDAvis: do the topic circles separate well, or do they overlap heavily?

- The circles do not overlap at all (which is surprising). Each circle is on opposite sides of the graph with no overlap.
- The topics are quite proportional with each topic holding about 20-30% of the tokens.

Interpret your topics:

Topic 1 focus on broad impacts of climate change on oceans and coastal environments. Words like "change", "ocean", and "warming" led me to this conclusion.

Topic 2 focus on coral reef ecosystems, specifically their biological and chemical dynamics. Words like "carbonate" and "fish" led me to this conclusion. "Analysis" and "data" suggest a research-oriented approach to studying coral reef ecosystems.

Topic 3 focus on coral bleaching and its effects on reef biodiversity and marine life. Words like "bleaching", "coral", and "species" led me to this conclusion. "Scientists" and "world" suggest a global scientific concern or international research attention surrounding coral reef decline.

Topic 4 appear to combine climate change, carbon, energy, and scientific communication . Words like "research", "change", and "carbon" led me to this conclusion.

Words that surprised me: fig, data, analysis, al, pdf. These metadata words should have been taken out at the beginning as they are irrelative to the thematic content. "Fig" is most likely due to figure captions. I originally thought "al" was "Al" miss-spelled. However, I think "al" refers to citations "et al". The thematically scientific words are not surprising.